

R-Based Policy Analysis Work Sample

Nusanta Samayel Audri | Incoming MPP Student, Fall 2026 | Public Policy / Applied Data Analysis

Project summary

Safety Nets Effect on Climate-Vulnerable Women’s Health: Internally Migrated from Northern Bangladesh

This sample summarizes the quantitative component of my master’s thesis, selected for publication through the Centre for Policy Dialogue Climate Compendium. The study examines whether social safety-net access and neighborhood-level clinic availability are associated with health and livelihood outcomes among climate-displaced women in three Dhaka informal settlements: Korail, Kallyanpur, and City Colony.

Dataset and design

- Primary survey dataset: 120 households and 342 individuals across three urban informal settlements.
- Sampling: 40 households per settlement, with proportional sub-area allocation based on field-mapped household estimates.
- Mixed-methods design: household survey, 30 in-depth interviews, 6 focus group discussions, and key informant interviews.
- Quantitative goal: estimate associations between safety-net receipt, clinic access, income, displacement type, health status, and post-migration employment transitions.

R workflow and data processing

- Cleaned and standardized raw survey responses; harmonized binary variables; converted income fields from text to numeric values; prepared model-ready analysis data.
- Constructed derived variables: good_health, higher_illness_burden, safety_net_access, clinic_access, income_after, displacement_type, and employment_transition.
- Produced descriptive tables, cross-tabulations, plots, and manuscript-ready regression tables; interpreted odds ratios and model fit.
- Conducted checks including multicollinearity screening, sensitivity checks using alternative health thresholds, and comparison of bivariate vs adjusted model results.

Core model specification

Main outcome: self-reported good health. Coefficients were interpreted as adjusted associations because the study uses cross-sectional observational data.

Good health = f (safety net access + clinic access + income after + displacement type)

Selected quantitative results

Model / Test	Key predictor	Result	Interpretation
Logistic regression	Clinic access	OR $\approx$ 3.34; $p \approx$ 0.013	Strongest adjusted predictor of good self-reported health.
Logistic regression	Safety-net receipt	n.s. after controls	Transfers alone did not predict health once structural mediators were included.
Employment transition model	Safety-net receipt	OR $\approx$ 2.54; $p \approx$ 0.0298	Associated with improved post-migration employment transition.
Bivariate test	Displacement type	$p \approx$ 0.029	Health vulnerability varied by displacement pathway.

Relevance to the RA position

- Directly demonstrates the minimum requirement: R-based data cleaning, processing, regression modeling, and quantitative reporting.
- Shows ability to construct an analysis dataset from messy field data and document variable operationalization clearly.
- Connects substantively to place-based inequality, service access, neighborhood conditions, and applied policy evaluation.
- Provides a foundation for the role's next-step methods: administrative data construction, data merging, quasi-experimental evaluation, and geospatial processing.

Best framing for the application

This sample should be submitted as evidence of R-based empirical workflow, not as evidence of causal inference expertise. The clean framing is:

"Although this thesis uses cross-sectional survey data rather than quasi-experimental administrative data, it reflects the core RA skills needed for empirical policy research: careful dataset construction, reproducible cleaning decisions, transparent variable coding, regression modeling, interpretation of estimates, and concise reporting for an academic manuscript."

How I would present my current skill level

- Strong: survey-data cleaning, variable construction, logistic regression, descriptive analysis, visualization, and policy interpretation.
- Developing: applied microeconometrics, difference-in-differences, regression discontinuity, and geospatial data workflows.
- Contribution fit: ready to support cleaning, merging, documentation, replication checks, descriptive analysis, and regression workflows while building more advanced causal and geospatial skills under supervision.

- **Selected R Workflow Excerpt**

The excerpt below summarizes the core structure of the analysis workflow. The full attached R script documents the complete process, including multi-sheet data import, cleaning, merging, variable construction, bivariate tests, logistic regression, odds-ratio visualization, and output export.

```
# Load and clean multiple survey sheets
demo <- read_excel(data_path, sheet = "Demographics") %>% clean_names()
ha <- read_excel(data_path, sheet = "Health Access") %>% clean_names()
hia <- read_excel(data_path, sheet = "Health Issues & Aid") %>% clean_names()
mig <- read_excel(data_path, sheet = "Migration Impact") %>% clean_names()

yn_clean <- function(x) {
  x %>%
    as.character() %>%
    str_trim() %>%
    str_to_lower() %>%
    case_when(
      . %in% c("yes", "y", "1", "true") ~ "Yes",
      . %in% c("no", "n", "0", "false") ~ "No",
      TRUE ~ NA_character_
    )
}

# Build model-ready analytic dataset
df <- ha %>%
  select(household_id, major_reason_of_displacement,
         years_staying_in_dhaka, disease_frequency_in_weeks,
         clinic_access) %>%
  left_join(hia_household, by = "household_id") %>%
  left_join(mig_clean, by = "household_id") %>%
  mutate(
    safety_net_binary = if_else(aid_received_on_health == "Never Received", "No", "Yes"),
    self_health = case_when(
      disease_frequency_in_weeks <= 1 ~ "Good",
      disease_frequency_in_weeks >= 3 ~ "Poor",
      TRUE ~ "Moderate"
    ),
    good_health = if_else(self_health == "Good", 1, 0),
    income_after_k = income_after / 1000,
    displacement_group = case_when(
      str_detect(str_to_lower(major_reason_of_displacement), "flood|erosion") ~ "Flood/Erosion",
      str_detect(str_to_lower(major_reason_of_displacement), "drought|crop|starvation") ~
"Drought/Crop Loss",
      TRUE ~ "Other/Social"
    )
  )

# Estimate adjusted logistic regression model
model_health <- glm(
  good_health ~ safety_net_binary + clinic_access + income_after_k + displacement_group,
  family = binomial(link = "logit"),
  data = df
)

# Export odds ratios and model visualization
or_table <- tidy(model_health, conf.int = TRUE, exponentiate = TRUE) %>%
  filter(term != "(Intercept)")

write_csv(or_table, "logistic_regression_odds_ratios.csv")
ggsave("adjusted_odds_ratios_good_health.png", width = 8, height = 5, dpi = 300)
```